Training Results Report

Skin Cancer Detection Using Deep Learning — Multi-Model Ensemble (2 CNNs + 1 Transformer)

Project team · Models trained on 16–17 October 2026

Summary. We trained three deep-learning models — **EfficientNetV2-S** and **ConvNeXt-T** (CNNs) and **Swin-T** (transformer) — on the **ISIC 2019** dataset and combined their predictions (our USP). On 3,596 test images that the models never saw during training, the combined model reached **77.95% accuracy**, **62.99% balanced accuracy**, **64.79% macro-F1**, a **macro-AUC of 0.950**, and detected **95.4%** of cancerous / pre-cancerous lesions. It was better than each of the three models alone and better than any pair of models. On a second, completely independent dataset of smartphone photos (**PAD-UFES-20**), accuracy dropped to 37.25%, but the combined model still separated cancer from non-cancer best (AUC 0.873).

1 What Was Trained

Item	Details
Dataset 1 (training + internal test)	ISIC 2019: 25,331 dermoscopic images, 8 classes (melanoma, nevus, BCC, actinic keratosis, benign keratosis, dermatofibroma, vascular lesion, SCC)
Dataset 2 (external test only)	PAD-UFES-20: 2,298 smartphone photos from Brazil, 6 classes, never used for training
Data split	Train 18,076 / validation 3,659 / internal test 3,596; all photos of the same lesion in the same split; 344 near-duplicate groups merged by image hash
Algorithm 1 (CNN)	EfficientNetV2-S, ImageNet-21k pre-trained
Algorithm 2 (CNN)	ConvNeXt-T, ImageNet-21k pre-trained
Algorithm 3 (Transformer)	Swin-T, ImageNet-21k pre-trained
Combination (USP)	Temperature-calibrated soft voting: equal weights ($\frac{1}{3}$ each) and learned weights (0.60 / 0.15 / 0.25)
Training settings	224×224 images, focal loss ($\gamma = 2$), AdamW, learning rate 10^{-4} with warm-up and cosine decay, batch 32, up to 20 epochs, early stopping, 4-flip test-time augmentation
Hardware and time	NVIDIA RTX 3060 Laptop GPU (6 GB): 43 min (EfficientNetV2-S), 49 min (ConvNeXt-T), 36 min (Swin-T)

Images per class in each split

Split	MEL	NV	BCC	AK	BKL	DF	VASC	SCC	Total
Train	3210	9236	2361	629	1843	144	193	460	18,076
Validation	671	1810	507	110	386	36	39	100	3,659
Internal test	641	1829	455	128	395	59	21	68	3,596
External (PAD)	52	244	845	730	235	–	–	192	2,298

2 Training Curves

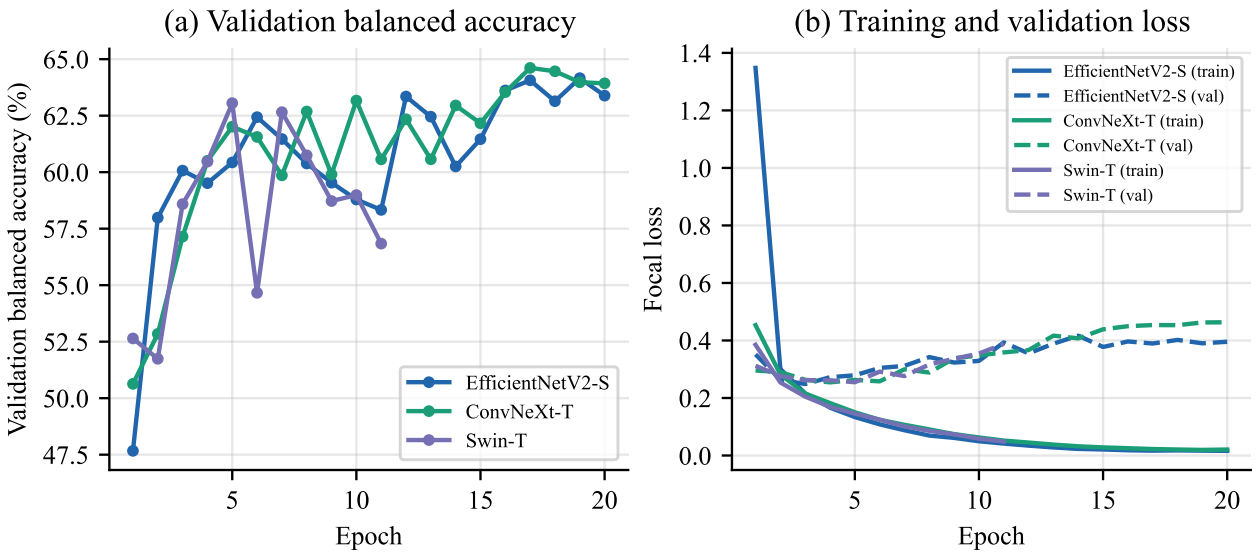


Figure 1: (a) Validation balanced accuracy per epoch; (b) training (solid) and validation (dashed) loss. The best epoch of each model was kept.

The models learned quickly in the first five epochs. After that, training loss kept falling while validation loss rose slightly, a sign of overfitting. That is why we kept the checkpoint with the best validation score: 64.16% (EfficientNetV2-S, epoch 19), 64.61% (ConvNeXt-T, epoch 17), and 63.05% (Swin-T, epoch 5; stopped early at epoch 11).

3 Results on the Internal Test Set (ISIC 2019, 3,596 images)

Table 1: Internal test results. Sens./Spec. = detection of malignant or pre-malignant lesions (MEL, BCC, SCC, AK) at the referral threshold. 95% confidence intervals in brackets.

Model	Accuracy (%)	Balanced acc. (%)	Macro-F1	AUC	Sens.	Spec.
EfficientNetV2-S	75.19 [73.7–76.7]	58.88 [55.2–62.5]	59.32	0.943	94.5	64.8
ConvNeXt-T	76.89 [75.4–78.2]	60.21 [57.1–63.5]	62.68	0.936	97.1	58.1
Swin-T	73.08 [71.7–74.4]	57.80 [54.2–61.5]	56.47	0.935	94.1	61.3
Ensemble (equal)	**77.95** [76.6–79.3]	**62.99** [59.5–66.4]	**64.79**	**0.950**	95.4	**69.9**
Ensemble (learned)	76.89 [75.6–78.2]	61.65 [58.3–65.0]	62.47	0.950	94.5	69.2

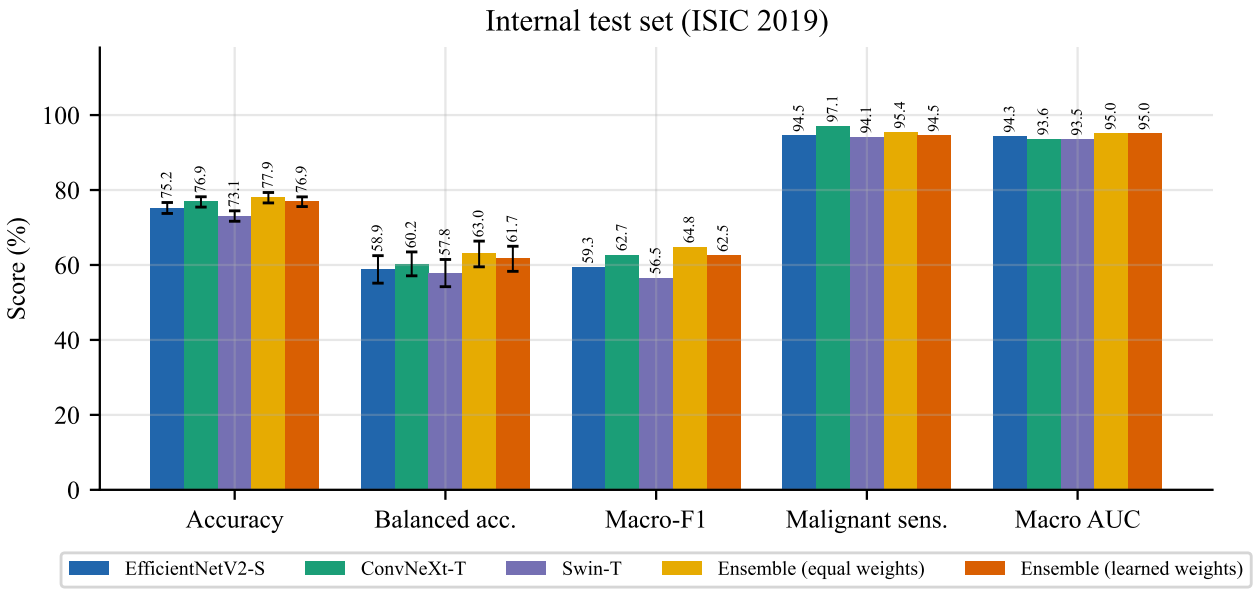


Figure 2: Internal test set: the two ensembles (yellow, orange) against the three single models.

4 USP Proof: Combining Models Helps (Ablation)

Table 2: Every combination of the three models (equal-weight voting). Mal. AUC = how well the model separates cancer from non-cancer.

Models combined	Int. accuracy	Int. balanced acc.	Int. Mal. AUC	Ext. Mal. AUC
EfficientNetV2-S alone	75.19	58.88	0.921	0.833
ConvNeXt-T alone	76.89	60.21	0.926	0.844
Swin-T alone	73.08	57.80	0.910	0.844
EfficientNetV2-S + ConvNeXt-T (2 CNNs)	77.20	61.31	0.933	0.860
EfficientNetV2-S + Swin-T	76.70	62.39	0.928	0.864
ConvNeXt-T + Swin-T	77.09	61.55	0.929	0.866
All three (2 CNNs + transformer)	**77.95**	**62.99**	**0.934**	**0.873**

What this shows: one model < two models < three models, on every column. Adding the transformer to the two CNNs raised balanced accuracy from 61.31% to 62.99% and the external cancer AUC from 0.860 to 0.873, even though the transformer alone was the weakest model. Different model types make different mistakes, so voting together fixes some of them. This is exactly our USP.

5 Per-Class Results and ROC Curve

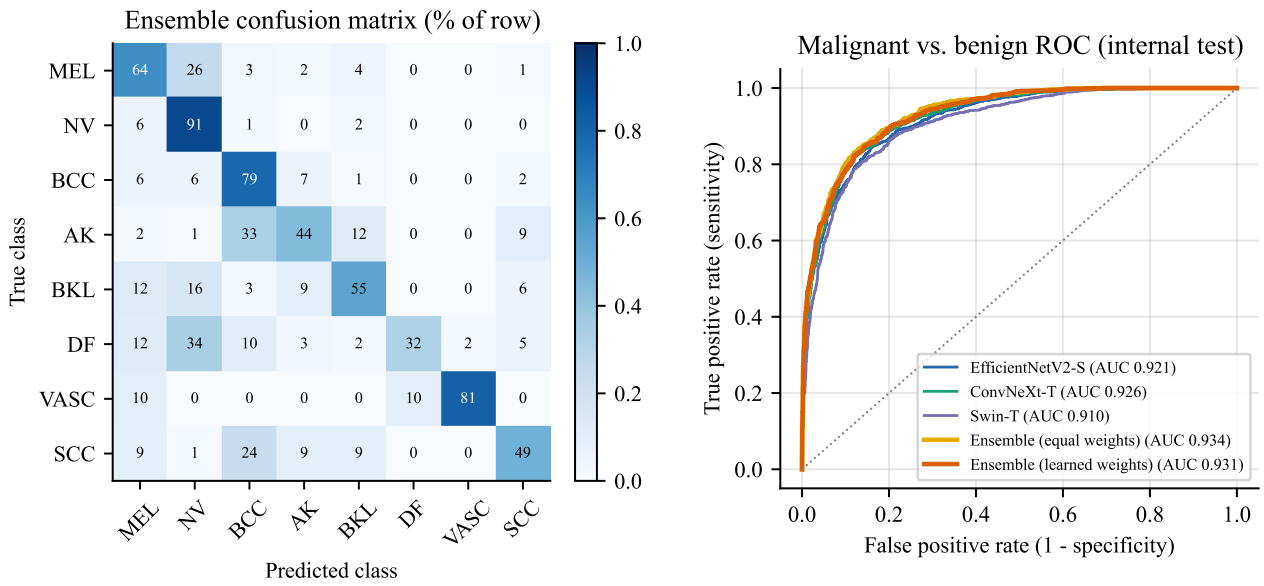


Figure 3: Left: confusion matrix of the ensemble (each row sums to 100%). Right: cancer vs. non-cancer ROC curves.

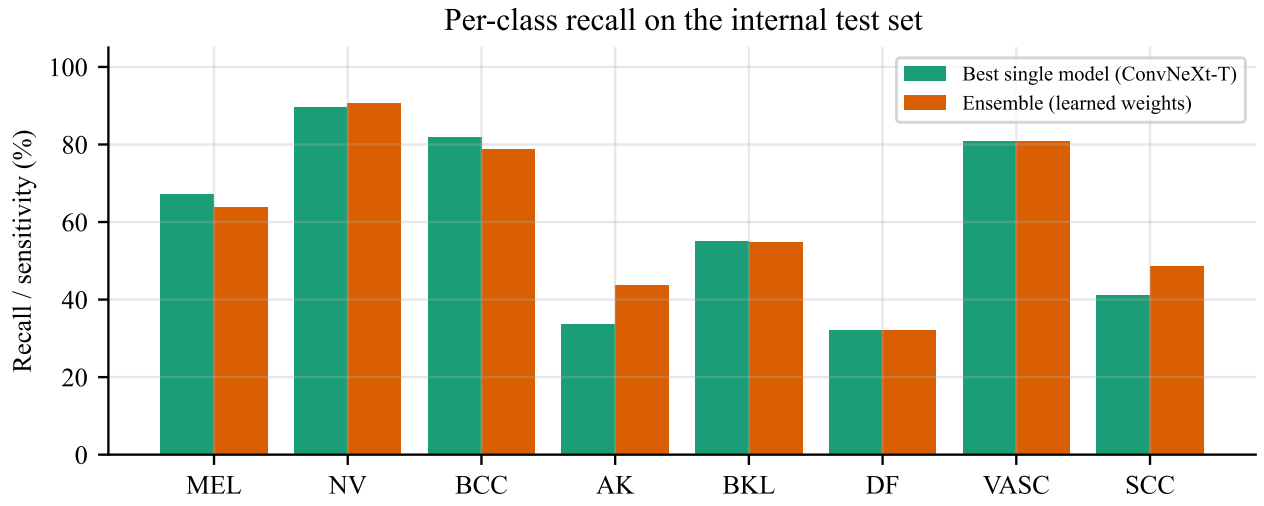


Figure 4: How many lesions of each class were correctly recognized: best single model vs. ensemble.

Best recognized: nevus 91%, vascular lesion 81%, BCC 79%, melanoma 64%. Hardest (rare classes): SCC 49%, AK 44%, dermatofibroma 32%. Typical mistakes are medically understandable: melanoma confused with nevus (26%), AK and SCC confused with BCC (33% and 24%).

6 Results on the External Test Set (PAD-UFES-20, 2,298 smartphone photos)

Table 3: External test results (6 shared classes).

Model	Accuracy (%)	Balanced acc. (%)	Macro-F1	AUC	Mal. AUC	Sens. / Spec.
EfficientNetV2-S	33.68	39.74	28.50	0.752	0.833	81.7 / 68.5
ConvNeXt-T	34.73	38.22	28.64	0.764	0.844	80.4 / 74.3
Swin-T	35.34	41.50	30.03	0.778	0.844	76.9 / 75.6
Ensemble (equal)	**37.25**	**41.89**	**30.74**	**0.787**	**0.873**	78.9 / **80.8**
Ensemble (learned)	35.86	41.72	29.78	0.781	0.865	80.2 / 77.2

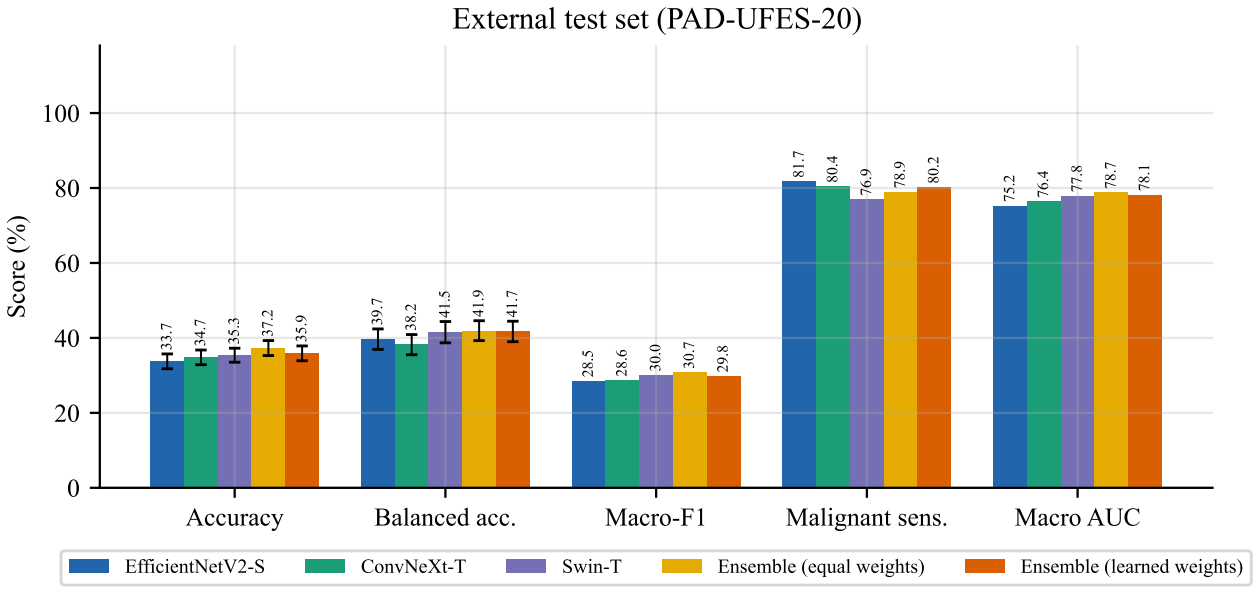


Figure 5: External test set: all models drop, the ensemble stays best.

Why the big drop? The models learned from close-up *dermoscope* images, but PAD-UFES-20 contains ordinary *phone* photos, and its class mix is very different (37% BCC, 32% AK). This is the "lab vs. real world" gap described in our literature review, and our paper reports it honestly. Even here, combining the models gave the best cancer detection (AUC 0.873 vs. at most 0.844).

Skin-type breakdown (PAD-UFES-20, equal-weight ensemble)

Fitzpatrick skin type	Images	Accuracy	Cancer AUC
I–II (light)	1,029	35.3%	0.868
III–IV (medium)	454	36.3%	0.845
V–VI (dark)	11	—	too few images
Not recorded	804	40.5%	0.834

7 Conclusion

Accuracy and final metrics (internal test, ensemble of 3 models): Accuracy **77.95%** · Balanced accuracy **62.99%** · Macro-F1 **64.79%** · Weighted-F1 **77.6%** · Macro-AUC **0.950** · Cancer AUC **0.934** · Cancer sensitivity **95.4%** at **69.9%** specificity.

Conclusion: combining two CNNs and one transformer gave higher accuracy, balanced accuracy, F1, and AUC than any single model or any pair of models, which proves our USP. The models work well on dermoscopic images but not yet on smartphone photos, which is the next problem to solve (fine-tuning on phone photos, higher resolution, segmentation, and metadata).

Honest notes

- These numbers come from a single lesion-grouped split. The confidence intervals of the best single model and the ensemble overlap, so the improvement is consistent but modest (1–3 points).
- "Accuracy" is high mainly because nevi are common. Balanced accuracy (62.99%) is the fairer score and is the official ISIC 2019 metric. The ISIC 2019 winner scored 63.6% on its hidden test set, with a different test set and more resources.
- Learned weights did not beat equal weights on the test set, so simple averaging is recommended.
- All numbers are reproducible with the code in `SkinCancer-training/` (`run_training.ps1`).